

Buddy drive MCU wiring – block-level harness (nets by label)

Source of truth: firmware/drive_mcu/docs/pin_map.md + include/pins.h;
generated by tools/kicad/gen_drive_wiring.py – regenerate, don't hand-edit.

PA2/PA3 = USART2 -> ST-LINK VCP -> Jetson USB (on-board, not wired here). ADR=0006.
Power: +12V_BUS via 60 A fuse from 3S pack (ADR=0005); +5V_RAIL buck powers Nucleo E5V (JP3=E5V);
+3V3 from Nucleo LDO feeds encoders + VN5019 logic. GND_PWR = motor return, star at battery.
ESTOP_STATE: contact off the E-stop chain, reporting only – E-stop cuts +12V_BUS upstream.
EN/DIAG lines need 3V3 pull-ups (VN5019 carrier has them); CS valid only while driving.

System-level power distribution and the Jetson/sensor interconnect are a separate
sheet: electronics/KICAD/system_wiring/ (tools/kicad/gen_system_wiring.py).

